ASSETS 2026

See What I Mean

A Technology Probe of AI Assistants for Chart Exploration with Low Vision

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Problem Magnification enlarges details but fragments the chart and loses the overview. AI chart assistants mostly serve screen reader users, yet low-vision individuals prefer using their **residual vision**.

RQ1 How do low-vision individuals experience AI-supported chart interaction grounded in their vision?

→ **Sharing their view helped** under both personas: it complemented vision, reduced effort, and kept independence.

RQ2 Should AI support systems for low-vision individuals adapt their response content?

→ **Adapted answers mattered**: 8 of 12 preferred the AI that adapts its answers to low vision, and 9 of 12 followed its directions.

Adapt both sides: what the AI sees and what it says

During exploration: ask, follow, point

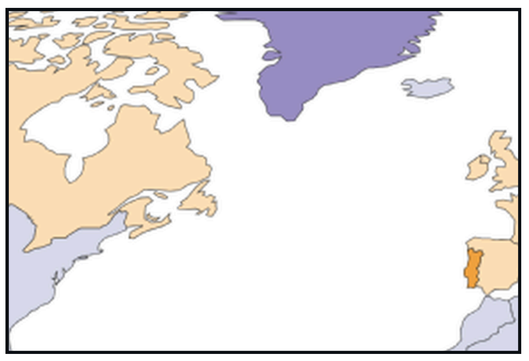
1 Ask



Which countries have the highest female population?

"...Outside your view to the right: Russia, Ukraine, Belarus, and Portugal."

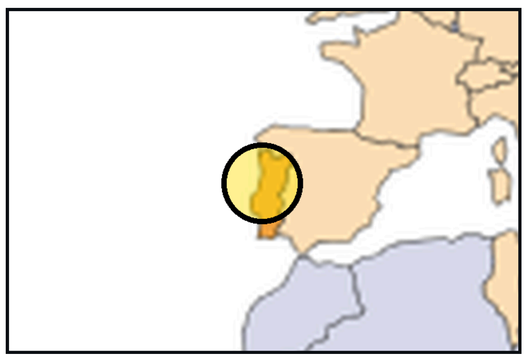
2 Follow guidance



Can you tell me how to find Portugal on this map?

"Portugal is visible on the right edge of your current view, about halfway down."

3 Point to confirm

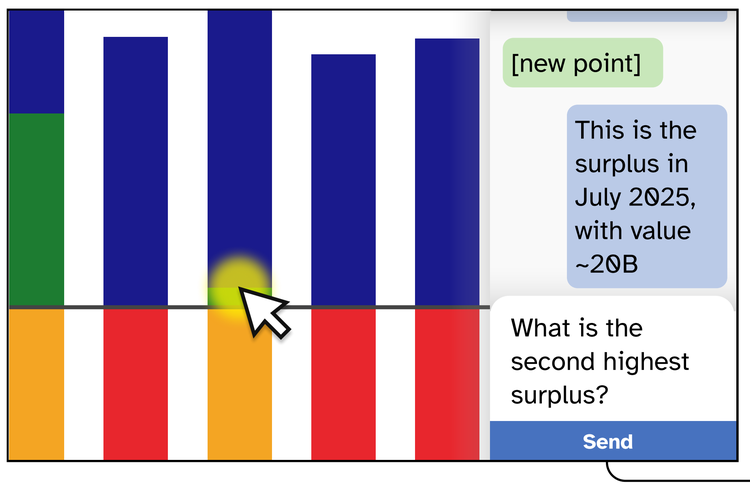


[double-clicks the country]

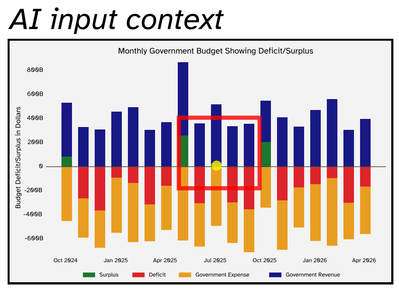
"You are pointing to Portugal. It is colored dark orange..."

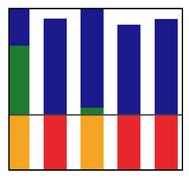
How the probe works: AI shares your view and interest point

Approach The AI assistant sees what you see: your **magnified view** and **point of interest**.



AI input context





Spatial
Semantic
Guidance

General AI (GAI)

The second highest surplus is 205B, in October 2025.

Low-vision AI (LVAI)

The second highest surplus is 205B, in October 2025. It is outside your view, to the right, about one viewport width. The green bar you are pointing is the smallest surplus, 20B, and the highest, 257B, is visible at the left edge of your view.

Technology probe study

12 low-vision participants explored two charts, one per AI persona: a **general AI (GAI)** and a **low-vision AI (LVAI)** that adapts its answers.

RQ1 · Sharing my view helped

Complement vision

Filled in **what vision missed**: colors, labels, and values out of view.

Agency and independence

Asked only when needed, without the social cost of asking a person.

Reduce effort

One click replaced panning, zooming, and cross-referencing.

89.6 **Usability (SUS)** out of 100

RQ2 · Adapted answers mattered

Spatial guidance in answers

LVAI	71.2%
GAI	7.8%

Semantic guidance in answers

LVAI	84.8%
GAI	29.1%

> "...much more specific... better suited to [their] personal needs." **P2**

> "I could click on it, it could tell me, and I could keep moving." **P12**

Limitation **5 of 12** found the LVAI **wordier** than they needed in the moment.

> "Sometimes I want an easy answer, value, name." **P1**

Design recommendations

DR1

Anchor answers on the chart
Pin, highlight, and annotate in the user's view.

DR2

Design for **verifiability**
Show what the AI refers to, so users can check it.

DR3

Scale answers to the action
Quick answers for clicks, detail for typed questions.

DR4

Augment vision, **keep agency**
Keep exchanges user-initiated; the user's view leads.

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